

Agniveer Vayu Other than Science (Group Y) - 25 July 2022 - Memory Based Paper

English

Question 1

Select the most appropriate synonym of the following given word:

Abhorrent

Options:

- A. Innocuous
- B. Inoffensive
- C. Attractive
- D. Awful

Answer: D

Solution:

The correct answer is "Awful"

Key Points

- Let's see the meanings of the given words-
 - **Abhorrent**-causing or deserving strong dislike or hatred
 - **Example**-Racism of any kind is abhorrent to me.
 - **Innocuous**-completely harmless
 - **Example**-Some mushrooms look **innocuous** but are in fact poisonous.
 - **Inoffensive**-not causing any harm or offense.
 - **Example**-He seemed like a quiet, **inoffensive** sort of a guy.
 - **Attractive**-very pleasing in appearance or sound
 - **Example**-I find him very **attractive**
 - **Awful**-extremely bad or unpleasant
 - **Example**-He suffered **awful** injuries in the crash.
- Hence, the correct answer is **Option 4 i.e. "Awful"**

Question 2

Out of the given four words only one is spelled correctly. Find the word that is correctly spelled.

Options:

A. Democratic

B. Suferred

C. Weaher

D. Horible

Answer: A

Solution:

The correct spelled word is "**Democratic**"



Key Points

- The correct spelling is "**Democratic**" which means "based on the principles of democracy"
 - Example-Do you think Australia is a more democratic country than Britain?
- Let's see the meanings of the other words-
 - **Suffered**-to experience physical or mental pain
 - **Weather**-the conditions in the air above the earth such as wind, rain, or temperature, especially at a particular time over a particular area
 - **Horrible**-very unpleasant or bad

Hence, the correct answer is **Option 1 i.e. Democratic.**

Question 3

Direction: Select the segment of the sentence that contains the grammatical error. If there is no error, mark 'No error' as your answer.

Hardly had he completed (A)/ his homework than the (B)/ lights went out. (C)/ No error (D)

Options:

A. A

B. B

C. C

D. D

Answer: B

Solution:

The correct answer is B.



Key Points

- A **conjunction** is a word that is used for **joining other words, phrases or sentences**.
- **Correlative conjunctions** are pairs of words that connect two parts of a sentence that have an equal value or correlate with each other.
 - Examples: **too...to, neither...nor, hardly...when, no sooner...than**, etc.
- We must use the **correct pair** of correlatives.
- The correct pair is '**hardly...when**' and not '**hardly...than**'.
 - **Example:** **Hardly** had I reached the station **when** the train left.
- So, in the given sentence, '**than**' should be replaced with '**when**' to make it grammatically correct.

Correct sentence: *Hardly had he completed his homework **when** the lights went out.*



Additional Information

- When a sentence begins with a **negative word** like '**not**', '**neither**', '**no sooner**', '**hardly**', '**scarcely**', etc., then the sentence will be in **inversion**, i.e., the **helping verb will come before the subject**.



Hinglish

- एक **conjunction** एक शब्द है जो अन्य **words, phrases** या **sentences** को जोड़ने के लिए उपयोग किया जाता है।
- **Correlative conjunctions** ऐसे शब्दों के जोड़े होते हैं जो एक वाक्य के दो हिस्सों को जोड़ते हैं जिनका एक समान मूल्य होता है या एक दूसरे के साथ सहसंबंधित होता है।
 - **उदाहरण:** **too...to, neither...nor, hardly...when, no sooner...than**, आदि।
- हमें सहसंबंधों की सही जोड़ी का उपयोग करना चाहिए।
- सही जोड़ी है 'मुश्किल से ... जब' और 'मुश्किल से ... नहीं'।
 - **उदाहरण:** **Hardly** had I reached the station **when** the train left.

- तो, दिए गए वाक्य में, 'than' को 'when' के साथ बदल दिया जाना चाहिए ताकि इसे व्याकरणिक रूप से सही किया जा सके।

Question 4

Identify the word that is opposite in meaning of the word 'INADVERTENT'.

Options:

- A. adequate
- B. available
- C. intentional
- D. negligent

Answer: C

Solution:

The correct antonym is 'Intentional'.



Key Points

- The word 'Inadvertent' means 'done or happening unintentionally (अनजाने में किया या हो रहा है)'.
- The antonyms of the word 'Inadvertent' are 'calculated, deliberate, intended, intentional, planned, premeditated, premeditative, premeditated, set'.
- From the antonym of the given word, we can say that the word 'Intentional' is the opposite in meaning.
- The word 'Intentional' means 'planned or intended (नियोजित या इरादा)'.

Hence, the correct answer is option 3. Let's see the meaning of other given options:-

Words	Meaning
Adequate	enough or satisfactory for a particular purpose (किसी विशेष उद्देश्य के लिए पर्याप्त या संतोषजनक)
Available	able to be bought or used (खरीदा या इस्तेमाल किया जा सकता है)
Negligent	not being careful or giving enough attention to people or things that are your responsibility (सावधान न रहना या लोगों या

उन चीज़ों पर पर्याप्त ध्यान न देना जो आपकी
ज़िम्मेदारी हैं)

Additional Information

- The **synonyms** of the word '**Inadvertent**' are 'accidental, casual, chance, fluky (also flukey), fortuitous, incidental, unintended, unintentional, unplanned, unpremeditated, unwitting'.
- Example of '**Inadvertent**' in a sentence:
 - *All authors need to be wary of **inadvertent** copying of other people's ideas.*

Question 5

Direction: Choose the appropriate word for the given sentence:

Your brother will be failed _____ he does not come tomorrow.

Options:

- A. if
- B. though
- C. unless
- D. until

Answer: A

Solution:

The correct answer is 'if.'

Key Points

- We often use **if** to introduce possible or impossible situations or conditions and their results.
- The situations or conditions can be real, imagined, or uncertain.
 - **Example:** *I usually make a sandwich to take to work **if** I have enough time.*
- In this sentence, the use of '**though**', '**unless**', and '**until**' would be incorrect.
 - **Though:** **Though** means '**in spite of something**.'
 - It is a subordinating conjunction.
 - This means that the clause which they introduce is a subordinate clause, which needs a main clause to make it complete.
 - When the **though** clause comes after the main clause, it can also mean '**but it is also true that ...**'

- **Example:** *We didn't make any profit **though** nobody knows why.*
- **Unless:** We use the conjunction **unless** to mean 'except if.'
 - **Example:** *They won't come **unless** you invite them.*
- **Until:** We use **until** as subordinating conjunction to connect an action or an event to a point in time.
 - **Example:** *We played chess **until** midnight.*
- Hence, the word, 'if' should be used to make the sentence grammatically correct.
- Hence, the correct answer is 'if.'
 - **The Correct Sentence:** *Your brother will be failed **if** he does not come tomorrow.*

Additional Information

- A **conjunction** is a joiner, a word that connects (conjoins) parts of a sentence.
- **Coordinating conjunctions** join equals to one another.
 - **Example:** *Most children like sweets **and** cookies.*
- Coordinating conjunctions go in between items joined, not at the beginning or end.
 - **Example:** *I like coffee, **but** I don't like tea.*

Hinglish

- हम अक्सर संभावित या असंभव स्थितियों या स्थितियों और उनके परिणामों को पेश करने के लिए **if** का उपयोग करते हैं।
- परिस्थितियाँ वास्तविक, काल्पनिक या अनिश्चित हो सकती हैं।
 - **Example:** *I usually make a sandwich to take to work **if** I have enough time.*
- इस वाक्य में 'though', 'unless' और 'until' का प्रयोग गलत होगा।
 - **Though:** **Though** का मतलब है 'कुछ के बावजूद।'
 - यह एक subordinating conjunction है।
 - इसका अर्थ यह है कि वे जिस clause का परिचय देते हैं वह एक subordinate clause है, जिसे पूरा करने के लिए एक main clause की आवश्यकता होती है।
 - जब main clause के बाद **though** उपवाक्य आता है, तो इसका अर्थ यह भी हो सकता है कि 'लेकिन यह भी सत्य है कि...'
 - **Example:** *We didn't make any profit **though** nobody knows why.*
 - **Unless:** हम conjunction **unless** का अर्थ 'except if' हैं।
 - **Example:** *They won't come **unless** you invite them.*
 - **Until:** हम **until** किसी क्रिया या घटना को समय पर एक बिंदु से जोड़ने के लिए एक subordinating conjunction के रूप में उपयोग करते हैं।
 - **Example:** *We played chess **until** midnight.*
- अतः वाक्य को व्याकरणिक रूप से शुद्ध करने के लिए 'if' शब्द का प्रयोग करना चाहिए।

Question 6

Direction: In the following question out of the four alternatives, choose the one which can be a substitute for the given word/sentence.

One who eats everything

Options:

- A. Omnivorous
- B. Omniscient
- C. Irresistible
- D. Insolvent

Answer: A

Solution:

The correct answer is '**Omnivorous.**'



Key Points

- One who eats everything is known as '**Omnivorous.**'
 - जो सब कुछ खाता है उसे 'सर्वभक्षी' ('**Omnivorous**') कहा जाता है।
 - **Example:** Both species are *omnivorous*, feeding voraciously on fruits and insects.
- Therefore, the correct answer is '**Omnivorous.**'



Additional Information

- Let us discuss the meanings of other options:

Word	Meaning	अर्थ
Omniscient	knowing everything.	सब कुछ जानना।
Irresistible	too attractive and tempting to be resisted.	विरोध करने के लिए बहुत आकर्षक।
Insolvent	unable to pay debts owed.	बकाया कर्ज चुकाने में असमर्थ।

Question 7

Select the most appropriate synonym for the given word.

Fury

Options:

- A. Crime
- B. Sin
- C. Anger
- D. Risk

Answer: C

Solution:

The correct answer is **Option 3**.

- **Fury:** wild or violent anger
- **Anger:** a strong feeling of annoyance, displeasure, or hostility.
- **Crime:** an action or omission which constitutes an offence and is punishable by law.
- **Sin:** an immoral act considered to be a transgression against divine law.
- **Risk:** a situation involving exposure to danger.

Anger is synonymous with **fury**.

Question 8

Select the most appropriate meaning of the given idiom.

Draw the line at something

Options:

- A. Coming to a conclusion
- B. Making pencil sketches
- C. Accept something up to a particular point
- D. Agreeing to an idea

Answer: C

Solution:

The correct answer is option 3) i.e., 'Accept something up to a particular point'

Key Points

- **Draw the line at something** is an idiom that means **to set a limit on what one is willing to do or accept**.
 - Example - I **draw the line** at a pizza with anchovies on it
 - So, **accept something up to a particular point** is the most appropriate meaning of the given idiom.
-

Question 9

Give one word for 'one who hates women'.

Options:

- A. Misanthrope
- B. Misogynist
- C. Feminist
- D. Effeminate

Answer: B

Solution:

The correct answer is **Misogynist**.

Key Points

- A **Misogynist** is **a man who hates or discriminates against women**.
 - Example - Sheila knew that it would be difficult to get a top-level position in a company run by a **misogynist**.

Hence, **Option 2** is the correct answer.

Additional Information

- A **Misanthrope** is a person who hates or avoids other people.
 - A **Feminist** is a person who supports equal rights and opportunity for all genders.
 - An **Effeminate** is a man who exhibits feminine behavior.
-

Question 10

Directions: Choose the appropriate preposition for the given sentence:

His path was beset _____ difficulties.

Options:

- A. with
- B. within
- C. among
- D. by

Answer: A

Solution:

The correct answer is '**with**'.



Key Points

- The given sentence is saying something about **someone's difficult path**.
- The use of the word "**beset**" in the sentence **indicates be covered or studded with**.
- Therefore, the most appropriate preposition to be filled in the blank is '**with**'.
- '**With**' means **having or possessing something**.
 - **Example:** *The plan was beset **with** difficulties from the beginning.*

Hence, the correct answer is **option 1**.

Complete Sentence: *His path was beset **with** difficulties.*



Additional Information

- Let us explore the other options:
 - '**Within**' means **inside something**.
 - '**Among**' means **situated more or less centrally in relation to several other things**.
 - '**By**' is used **for identifying the agent performing an action**.
-

Question 11

What is the antonym of the word given below?

Initiate

Options:

- A. Start
- B. Begin
- C. Commence
- D. Finish

Answer: D

Solution:

The correct answer is **Finish**.



Key Points

Initiate:to cause something to begin

Finish:bring (a task or activity) to an end; complete

Thus, we can see that Initiate and Finish are antonyms.



Additional Information

The meaning of the other words:

- **Start:**to begin doing something
 - **Begin:**to start to do something
 - **Commence:**to begin something especially a shop or business.
 - **Initiate:** to take first step in doing something especially something new.
Start, Begin, Commence and Initiate are all near-synonyms.
-

Question 12

Select the correctly spelt word.

Options:

- A. Attendance
- B. Attendence
- C. Attandence
- D. Attandance

Answer: A

Solution:

The correct answer is **option 1), i.e. Attendance**.



Key Points

- The correct spelling of the word is '**Attendance**'.
 - The meaning of the word '**Attendance**' is 'the action of being present'. (उपस्थित होने की क्रिया)
-

Question 13

Read the sentence to find out whether there is any error in it. The error, if any, will be in one part of the sentence. The number of that part is the answer. If there is no error, the answer is (5). Ignore errors of punctuation, if any.

(1)How much boys/(2) were present/(3)in the assembly./(4) No Error

Options:

A. 1

B. 2

C. 3

D. 4

Answer: A

Solution:



Key Points

- Part 1 contains an error.
- We know that in case of uncountable nouns like milk, water, ,rice etc. we use **much** and not **many** but when the uncountable noun is supported by a countable noun like people, packet, bottle, bag etc. we should use **many** and not **much**.
- The question should start as *How many boys*.

Correct Sentence:How many boys were present in the assembly.

Question 14

Identify the segment in the sentence which contains the grammatical error.

We went for swimming into the river on Sunday.

Options:

- A. We went for swimming
- B. into the
- C. river on
- D. Sunday

Answer: B

Solution:

Option 2 is the right answer. It has an error.

A preposition is always **followed by an object**, which is a noun and never a verb. [subject +verb+ preposition+ object]

It joins a noun, a pronoun, or noun phrases to the rest of the sentence **to show their relation**.

'**into**' shows motion. However, in the given sentence, **a location** is indicated ' **a river**'.

The correct preposition would be '**in**', which indicates '**the location of the action**'.

The correct sentence, thus, is --**We went for swimming in the river on Sunday.**

Question 15

Direction: In the following question, out of the four alternatives, select the alternative which will improve the underlined part of the sentence. In case no improvement is needed, select “No improvement”.

No sooner Riya reached the station than she met her friend.

Options:

- A. Did Riya reached
- B. Did Riya reach
- C. Riya did reached
- D. No improvement

Answer: B

Solution:

The correct answer is “Did Riya reach”



Key Points

- In the given sentence ‘no sooner’ is used at the beginning of the sentence, we used inverted word order, i.e. Auxillary verb comes after no sooner and before the subject. Another example of inverted word order is: 'nosooner did I reach the station than the train had already left'.
- If the sentence starts with the subject and not with no sooner than there is no need to use an auxiliary verb, for example, a given sentence can also be written as “Riya no sooner reached the station than she met her friend.” this is also correct and conveys the same meaning.

Correct sentence: “No sooner did Riya reach the station than she met her friend.”

Question 16

Direction: Change the Narration-

“Whatareyoudoing?” He said to me

Options:

- A. He asked me what I was doing.
- B. He asked me what I were doing.
- C. He asked me that what I was doing.

D. He asked me what he was doing.

Answer: A

Solution:

The correct answer is '**He asked me what I was doing.**'

Key Points

- An **interrogative sentence** has a question.
- The rule for changing Direct Speech to Indirect Speech:
 - The given sentence is in **Present Continuous Tense**.
 - **Present Continuous Tense** changes to **Past Continuous tense**.
 - Remove the commas and inverted commas.
 - Do not use **that** in the sentence of indirect speech.
 - The reporting verb **said/said to** is changed in **asked, demanded, ordered, enquired** as per the nature of the sentence.
 - Remove the question mark in the sentence of Indirect speech.
 - There are some questions that cannot be answered using '**YES/NO.**'
 - To make the indirect speech of such questions, '**if/whether**' will not be used.
 - The question is changed to assertive form and is simply placed after the reporting verb.
 - **Example:** *The boy said to Ram, "What is your name?"* (**Direct speech**)
The boy asked Ram what his name was. (**Indirect speech**)
- Therefore, the correct answer is '**He asked me what I was doing.**'

Additional Information

- If the sentence of Direct speech has a question that cannot be answered by saying yes or no then it does not use "**if**" or "**whether**" in its Indirect speech.
 - **Example:** *She said to me, "Where did you buy this dress from?"* (**Direct speech**)
She asked me where I had bought that dress from. (**Indirect speech**)

Hinglish

- **Interrogative** वाक्य में एक प्रश्न होता है।
- Direct Speech को Indirect Speech में बदलने का नियम:
 - दिया गया वाक्य **Present Continuous Tense** में है।
 - **Present Continuous Tense, Past Continuous tense** में बदलता है।
 - Commas और inverted commas हटा दे।
 - Indirect speech के वाक्य में **that** प्रयोग न करें।
 - **Said/said to** रिपोर्टिंग verb वाक्य की प्रकृति के अनुसार **asked, demanded, ordered, enquired** में बदल जाती है।
 - Indirect speech के वाक्य में प्रश्न चिह्न हटा दें।
 - कुछ ऐसे प्रश्न हैं जिनका उत्तर '**YES/NO**' में नहीं दिया जा सकता है।
 - ऐसे प्रश्नों को Indirect speech देने के लिए '**if/whether**' का प्रयोग नहीं किया जाएगा।
 - Question को assertive form में बदल दिया जाता है और केवल reporting verb के बाद रखा जाता है।
 - **Example:** *The boy said to Ram, "What is your name?"* (**Direct speech**)
The boy asked Ram what his name was. (**Indirect speech**)

Question 17

The abstract noun of 'patient' is _____.

Options:

- A. Illness
- B. Patience
- C. Disease
- D. Sickness

Answer: B

Solution:

The correct answer is- '**Patience**'



Key Points

- '**Patient**' is a **common noun**.
- An **abstract noun** is the name of feeling, quality or condition, etc.
- Some **abstract nouns** are formed by removing the last letters and adding a suffix to the common noun.
- '**Patient**' is such a **common noun**.
- Thus the correct answer is: '**Patience**'



Additional Information

- Let's consider the usage of other words given in the options:
 - **Illness**: It is the abstract noun of ill.
 - **Disease**: It is a common noun.
 - **Sickness**: It is the abstract noun of sick.

Question 18

People in the villages of Rajasthan lead a very simple life. Their way of living has not changed over the years. They live in circular huts. The walls of these huts are covered with cowdung. Every hut has a small place for worship. The life of these people is full of difficulties. It is very hot in summers and cold in winters. Water is a major problem. Sometimes they have to walk a long distance to get drinking water. For their agriculture they depend on rains. But these people are very brave. They have learnt to face difficulties and they never lose hope.

They also like to enjoy their life. Women like to wear dresses full of bright colours. People living in villages in Rajasthan have a rich tradition of music and dance. The people of Rajasthan are very proud of their culture.

Which of the following is an adjective formed from the noun ‘music’?

Options:

- A. musician
- B. musical
- C. musically
- D. musicality

Answer: B

Solution:

The root word is ‘**music**’ (noun), it is the pattern of sounds produced by people singing or playing instruments.

- ‘**Musical**’ is an **adjective form of music**, it is something is connected with playing or studying music; something having a pleasant sound; melodious or tuneful.
- **Musician** (noun): A person who plays a musical instrument.
- **Musically** (adverb): In a way that relates to music.
- **Musicality** (noun): It is a talent or skill and understanding in performing music.

Hence, the word ‘**Musical**’ is an **adjective** formed from the **noun** ‘**Music**’.

Question 19

People in the villages of Rajasthan lead a very simple life. Their way of living has not changed over the years. They live in circular huts. The walls of these huts are covered with cowdung. Every hut has a small place for worship. The life of these people is full of difficulties. It is very hot in summers and cold in winters. Water is a major problem. Sometimes they have to walk a long distance to get drinking water. For their agriculture they depend on rains. But these people are very brave. They have learnt to face difficulties and they never lose hope. They also like to enjoy their life. Women like to wear dresses full of bright colours. People living in villages in Rajasthan have a rich tradition of music and dance. The people of Rajasthan are very proud of their culture.

Which of the following word is a synonym for ‘brave’?

Options:

- A. hardworking

B. valiant

C. prosperous

D. enthusiastic

Answer: B

Solution:

A synonym is a word or phrase that means exactly or nearly the same as another word or phrase in the same language.

Meaning of Brave: Courageous, showing no fear, daring

‘Valiant’ is the synonym of the word ‘Brave’.

- **Hardworking:** tending to work with energy and commitment
- **Prosperous:** fortunate financially, flourishing
- **Enthusiastic:** showing great excitement or interest

Hence, a synonym for the word ‘Brave’ is ‘Valiant’.

Question 20

People in the villages of Rajasthan lead a very simple life. Their way of living has not changed over the years. They live in circular huts. The walls of these huts are covered with cowdung. Every hut has a small place for worship. The life of these people is full of difficulties. It is very hot in summers and cold in winters. Water is a major problem. Sometimes they have to walk a long distance to get drinking water. For their agriculture they depend on rains. But these people are very brave. They have learnt to face difficulties and they never lose hope. They also like to enjoy their life. Women like to wear dresses full of bright colours. People living in villages in Rajasthan have a rich tradition of music and dance. The people of Rajasthan are very proud of their culture.

‘The way of living’ can be replaced with the word

Options:

A. livelihood

B. liveliness

C. lifelike

D. lifestyle

Answer: D

Solution:

‘**The way of living**’ means the particular way that a person lives. It is the way you live including your style, attitudes and possessions.

- **Lifestyle**(noun): The way in which a person lives
- **Livelihood**(noun): The money you need to pay for food, place to live, clothing, etc.
- **Liveliness**(noun): The quality of being interesting and exciting.
- **Lifelike**(adjective): Accurately representing or imitating real life.

Hence, ‘**The way of living**’ can be replaced with the word ‘**Lifestyle**’.

RAGA

Question 21

_____, built in the 11th century, is dedicated to Lord Shiva and is considered as the largest temple of the city Bhubaneswar (Odisha).

Options:

- A. Lingaraj Temple
- B. Sun Temple
- C. Badrinath Temple
- D. Brihadeeswara Temple

Answer: A

Solution:

The correct answer is Lingaraj Temple.



Key Points

- **Lingaraja Temple** is a **Hindu temple** dedicated to **Shiva** and is **one of the oldest temples** in **Bhubaneswar**, the **capital** of the state of **Odisha**.

- It was **built by King Jajati Keshari in the 10th Century** and **completed by King Lalatendu Keshari in the 11th Century**.
- The temple represents the **quintessence of Kalinga architecture** and culminates in the **medieval stages of the architectural tradition** at Bhubaneswar.

Additional Information

- **Konark Sun Temple** is a **13th-century CE** Sun temple at Konark, **near Puri city, Odisha**.
 - The temple is attributed to **king Narasimhadeva I of the Eastern Ganga dynasty** about **1250 CE**.
 - **Badrinath Temple** is a **Hindu temple dedicated to Vishnu** situated in the town of **Badrinath in Uttarakhand**.
 - The temple is also **one of the 108 Divya Desams** dedicated to Vishnu, who is worshipped as **Badrinath, holy shrines for Vaishnavites**.
 - **Brihadishvara Temple** is a **Hindu Dravidian-styled temple** dedicated to the **god Shiva** located on the **South bank of the Kaveri river in Thanjavur, Tamil Nadu**.
 - The temple has a **massive colonnaded Prakara (corridor)** and is **one of the largest Shiva lingas in India**.
 - It is also famed for the **quality of its sculpture**, as well as being the **location that commissioned the brass Nataraja – Shiva as the lord of dance, in the 11th century**.
-

Question 22

Which city is scheduled to host the Summer Olympic Games 2024?

Options:

- A. Paris
- B. Tokyo
- C. Los Angeles
- D. Beijing

Answer: A

Solution:

The correct answer is **Paris**.

Key Points

- The **2024 Summer Olympics** is a **forthcoming international multi-sport event** that is **scheduled to take place from 26 July to 11 August 2024** in **Paris, France**.
- **Paris** will become the **second city** to host the **Olympics** three times, after **London (1908, 1948, and 2012)**.

- The Olympic edition of 1900 and 1924 was hosted by Paris.
- The 2024 Games will mark the third time Paris has staged the Olympics, coming exactly 100 years after they last hosted the Games.

Additional Information

- The motto of 2024 Paris Games is **Made for Sharing**.
 - **2022 Winter Olympics** will be held in **Beijing**.
 - The official Mascot of the 2022 Winter Olympics Games is **Bing Dwen Dwen**.
 - 2020 Summer Olympics rescheduled to 2021 will be held in **Tokyo, Japan**.
 - The official Mascot of the 2020 Summer Olympics Games is **Miraitowa**.
-

Question 23

Who among the following is the winner of the US Open 2021 Men's Singles Final?

Options:

- A. Novak Djokovic
- B. Daniil Medvedev
- C. Rafael Nadal
- D. Roger Federer

Answer: B

Solution:

The correct answer is **Daniil Medvedev**.

Key Points

- Russia's **Daniil Medvedev** was crowned champion of the **2021 US Open** after defeating world number one **Novak Djokovic** in the men's singles final.
- He clinched his maiden Grand Slam title with a straight-set victory against the Serbian (6-4, 6-4, 6-4).
- Medvedev has become the third Russian after **Yevgeny Kafelnikov** and **Marat Safin** to win a Grand Slam.



Additional Information

- Britain's Emma Raducanu became the first qualifier to capture a Grand Slam title by defeating Canada's 19-year-old Leylah Fernandez in the **2021 US Open women's final**.
-

Question 24

When is the Indian Air Force day celebrated?

Options:

- A. 4 december
- B. 8 October
- C. 15 January
- D. 26 July

Answer: B

Solution:

The correct answer is **8 October**.

- The Indian Air Force day is celebrated on 8 October.

Key Points

- **Indian Air Force:**
 - Air Force day is celebrated on **8th October every year**.
 - Indian Air Force was officially established on **8 October 1932** as an auxiliary air force of the British Empire.
 - After India gained independence from the United Kingdom in 1947, the name **Royal Indian Air Force was kept till 1950** and after that the prefix Royal was removed.
- Since **October 8, 1932**, the day is celebrated every year as Air Force day

- **Air Chief Marshal Vivek Ram Chaudhari** is the current **Chief of Air Staff**.
 - He is the **27th Chief of the Air Staff**.

Additional Information

- **Indian Navy Day** is celebrated on **4 December**.
 - **Indian Army Day** is celebrated on **15 January**.
 - **Kargil Vijay Diwas** is celebrated on **26 July**.
-

Question 25

When the posts of both the President and the Vice President are vacant who among the following officiates as President?

Options:

- A. The Prime Minister
- B. The Speaker of Lok Sabha
- C. The Chief Justice of Supreme Court
- D. The Lt. Governor of Delhi

Answer: C

Solution:

The correct answer is **The Chief Justice of India**.

- **The Chief Justice of India** officially the **Chief Justice of the Supreme Court of India**, is the chief judge of the Supreme Court of India as well as the highest-ranking officer of the Indian federal judiciary.
- The Constitution of India grants power to the president of India to appoint, in consultation with the outgoing chief justice, the next chief justice, who will serve until they reach the age of sixty-five or are removed by impeachment.

Important Points

1. **The president (Discharge of Functions) Act, 1969 of India** provides that the chief justice of India shall act as the president of India in the event of the offices of both the president and the vice president being vacant.
2. **When President Zakir Hussain died in office, Vice President V. V. Giri, acted as the president. Later, Giri resigned as the vice president. The chief justice, Justice Mohammad Hidayatullah then became the acting president of India.**
3. As per the convention, the senior-most judge of the Supreme Court became the acting chief justice.

4. **Example-** *Mohammad Hidayatullah was the 11th Chief Justice of India.* He had also served as the *Acting President of India from 20 July 1969 to 24 August 1969 and from 6 October 1982 to 31 October 1982. In May 1969 when president Dr Zakir Hussain died, the then vice-president V.V.*

Hence, When the posts of both the president and the vice president are vacant, The Chief Justice of India officiates as President.

Question 26

Who came to India during the time of Chandragupta Maurya?

Options:

- A. Faxian (Fa Hien)
- B. Xuanzang (Hiuen Tsang)
- C. Megasthenes
- D. Strabo

Answer: C

Solution:

The correct answer is Megasthenes.

Key Points

- At the time of the treaty between the Greek ruler **Seleucus I Nicator** and the Indian ruler **Chandragupta Maurya**, he appears to have been serving as an officer under Sibyrtius.
- **Megasthenes** was an ancient Greek historian, diplomat and Indian ethnographer, and explorer in the Hellenistic period.
- **Megasthenes** wrote the book “**Indica**”.
- **Megasthenes** was a Greek ambassador of Seleucus I Nicator in the court of Chandragupta Maurya.

Additional Information

- **Huien Tsang** visited India in 630 A.D.- 645 A.D. He was a Chinese traveler. He visited India during the supremacy of Harsha Vardhana.
 - **AbdurRazzak** visited India in 1443 A.D.- 1444 A.D. He was also an ambassador of Persia. He visited India during the rule of Deva Raya II of Vijayanagar.
 - **Captain William Hawkins** visited India in 1608 A.D. – 1611 A.D. He visited India during the reign of Jahangir.
 - Captain William Hawkins led the first expedition of the English East India Company to India in 1609.
-

Question 27

Full form of CNG is _____.

Options:

- A. Combined Natural Gas
- B. Compressed Natural Gas
- C. Clean Natural Gas
- D. Carbon Natural Gas

Answer: B

Solution:

The correct answer is Compressed Natural Gas.



Key Points

- **The major component of bio-gas and Compressed Natural Gas (CNG) is Methane.**
- **Bio-CNG contains about 92-98 % of methane** and only 2-8 % carbon dioxide.
 - The high methane content and calorific value combined with the low quantity of moisture, hydrogen sulphide and impurities makes Bio-CNG an ideal fuel for automobiles and power generation.
- **Compressed natural gas** is a fuel that can be used in place of gasoline, diesel fuel and liquefied petroleum gas (LPG).
 - CNG is made by compressing natural gas, which is mainly composed of methane (CH₄).



Additional Information

Fuel Gases	Major component	Uses
Compressed natural gas (CNG)	Methane Ethane	• Can be used in place of gasoline, diesel fuel and liquefied petroleum gas (LPG)
Liquefied petroleum gas (LPG)	Butane Propane	• LPG is used for cooking. • An alternative to electric heating, heating oil, or kerosene.

- | | | |
|--|--|--|
| | | <ul style="list-style-type: none"> As it is odourless so Methyl Mercaptan is added to it so there may be a smell if LPG leaks from its storage container. |
|--|--|--|
-

Question 28

The value of $(256)^{0.16} \times (256)^{0.09}$ is

Options:

- A. 5
- B. 16
- C. 4
- D. 25

Answer: C

Solution:

Formula used:

$$a^m \times a^n = a^{m+n}$$

$$(a^m)^n = a^{mn}$$

Calculation:

$$(256)^{0.16} \times (256)^{0.09}$$

$$\text{Since, } a^m \times a^n = a^{m+n}$$

$$\Rightarrow 256^{0.16+0.09} = 256^{0.25}$$

$$\Rightarrow 256^{1/4} = (16^2)^{1/4}$$

$$\text{As we know } (a^m)^n = a^{mn}$$

$$\Rightarrow 16^{1/2} = (4^2)^{1/2}$$

$$\therefore (256)^{0.16} \times (256)^{0.09} = 4$$

Question 29

If Anuj walks at the speed of 4 km/hr, then he reaches his school 6 minutes late but if he walks at the speed of 5 km/hr, then he reaches 6 minutes before the scheduled time. What is the distance (in km) of his school from his house?

Options:

A. 4

B. 3

C. 6

D. 3.5

Answer: A

Solution:

Shortcut Trick

Formula used:

Required distance is given by

$$D = \frac{S_1 \times S_2}{S_1 - S_2} \times (t_1 - t_2)$$

Whichever is greater can be taken first from S_1 and S_2 same for t_1

and t_2 greater one is taken first in subtraction

Calculation:

Let the correct time be 't'

$$\Rightarrow D = \frac{(4 \times 5)}{(5 - 4)} \times \frac{(t + 6) - (t - 6)}{60} \quad (\because 1 \text{ min} = 1/60 \text{ hr})$$

$$\Rightarrow D = 20 \times 12/60 = 4$$

∴ The required distance is 4 Km.



Alternate Method

Let the distance between school from house is D

Actual time at which he must have reached school = t in minutes

According to the question,

$$4 = D/(t + 6)$$

$$\Rightarrow D = 4(t + 6) \text{ ----(i)}$$

And,

$$\Rightarrow D = 5(t - 6) \text{ ----(ii)}$$

From eq. (i) and (ii)

$$4t + 24 = 5t - 30$$

$$t = 54 \text{ minutes}$$

$$\text{Distance, } D = [4 \times (54 + 6)]/60 = 4 \text{ km}$$

∴ The required distance is 4 Km.

Question 30

If $\sin^4 x + \sin^2 x = 1$ then what is the value of $\cot^4 x + \cot^2 x$?

Options:

A. -2

B. -1

C. 2

D. 1

Answer: D

Solution:

Formula used:

- $1 - \sin^2\theta = \cos^2\theta$
- $\cos\theta/\sin\theta = \cot\theta$
- $\operatorname{cosec}\theta = 1/\sin\theta$
- $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

Calculation:

Given that

$$\sin^4x + \sin^2x = 1$$

$$\Rightarrow \sin^4x = 1 - \sin^2x$$

$$\Rightarrow \sin^4x = \cos^2x \text{ ---(1)}$$

Now the required value

$$\Rightarrow \cot^4x + \cot^2x$$

$$\Rightarrow \cot^2(\cot^2 + 1)$$

$$\Rightarrow \cot^2x \operatorname{cosec}^2x [\because 1 + \cot^2\theta = \operatorname{cosec}^2\theta]$$

$$\Rightarrow \frac{\cos^2x}{\sin^2x \sin^2x}$$

$$\Rightarrow \frac{\cos^2x}{\sin^4x} = \frac{\cos^2x}{\cos^2x} = 1 \text{ [From eq (1)]}$$

$\therefore$ The value of $\cot^4x + \cot^2x$ is 1.

Question 31

1 litre of water is added to 5 litres of alcohol-water solution which is containing 40% alcohol strength. The strength of alcohol in the new solution will be

Options:

A. 30%

B. 33%

C. $33\frac{1}{3}\%$

D. $33\frac{2}{3}\%$

Answer: C

Solution:

Given:

1 litre of water is added to 5 litres of alcohol-water solution containing 40% alcohol strength.

Calculations:

Alcohol in original solution = $(40/100) \times 5 = 2$ litres

Water in original solution = 3 litres

On adding 1-litre water, water becomes 4 litres.

Now, 6 litres of solution contains 2 litres of alcohol.

So 100 litres of solution contains

$$(2/6) \times 100 = 100/3 \%$$

$\therefore$ the correct answer is $33\frac{1}{3}\%$.

Question 32

The length of a rectangle is twice its breadth. If its length is decreased by 5 cm and breadth is increased by 5 cm, the area of the rectangle is increased by 75 sq cm. What is the length of the rectangle?

Options:

A. 18 cm

B. 16 cm

C. 40 cm

D. 20 cm

Answer: C

Solution:

Given

The length of a rectangle is twice its breadth.

Length is decreased by 5 cm and breadth is increased by 5 cm.

Area of the rectangle is increased by 75 cm^2

Concept used

Area of rectangle = Length $\times$ Breadth

Calculation

Length is twice the breadth of the rectangle.

Length = $2x$ & breadth = x

Area = $2x^2$ -----(1)

Length is decreased by 5 cm and breadth is increased by 5 cm.

Area is increased by 75 cm^2 .

$$\Rightarrow (2x - 5)(x + 5) = 2x^2 + 75$$

$$\Rightarrow 2x^2 + 10x - 5x - 25 = 2x^2 + 75$$

$$\Rightarrow 5x = 100$$

$$\Rightarrow x = 20$$

$$\therefore \text{Length of rectangle} = 2 \times 20 = 40 \text{ cm}$$

Question 33

Three cubes of sides 1 cm, 6 cm and 8 cm are melted to form a new cube. Find half of the surface area of the new cube?

Options:

A. 293 cm^2

B. 463 cm^2

C. 486 cm^2

D. 243cm^2

Answer: D

Solution:

Given:

Ratio of sides = 1: 6: 8

Formula used:

The volume of cube = $(\text{Side})^3$

Surface area of cube = $6(\text{side})^2$

Calculation:

Volume of new cube = Sum of the volume of three small cube

Since, the volume of cube = $(\text{Side})^3$

$$\Rightarrow \text{Volume of new cube} = 6^3 + 8^3 + 1^3$$

$$\Rightarrow \text{Volume of new cube} = 729 \text{ cm}^3$$

Since, the side of cube = $(\text{volume})^{1/3}$

$$\Rightarrow \text{Side of new cube} = (729)^{1/3} = 9 \text{ cm}$$

We know that, the surface area of cube = $6(\text{side})^2$

$$\Rightarrow \text{Surface area of new cube} = 6 \times 9^2 = 486 \text{ cm}^2$$

$$\therefore \text{Half of the surface area} = 243\text{cm}^2$$

Question 34

Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. The probability of drawing two aces is.

Options:

A. $\frac{12}{13}$

B. $\frac{15}{26}$

C. $\frac{9}{13}$

D. $\frac{1}{221}$

Answer: D

Solution:

Given:

A total playing card is 52, and the total Ace is 4

Concept used:

Probability = Number of favourable outcomes/Total number of Possible outcomes

Calculations:

Number of favorable outcomes is 4 Ace cards

Total number of playing cards is 52 cards

$$\text{Probability} = (4/52) \times (3/51)$$

$$\Rightarrow (1/13) \times (1/17) = 1/221$$

$\therefore$ The probability of drawing an ace from a deck of cards is 1/221.

Question 35

A ladder leaning against a wall makes an angle of 60° with the horizontal. If the foot of the ladder is 2.5 m away from the wall, find the length of the ladder.

Options:

A. $5/\sqrt{3}$ m

B. 5 m

C. 3 m

D. $3\sqrt{3}$ m

Answer: B

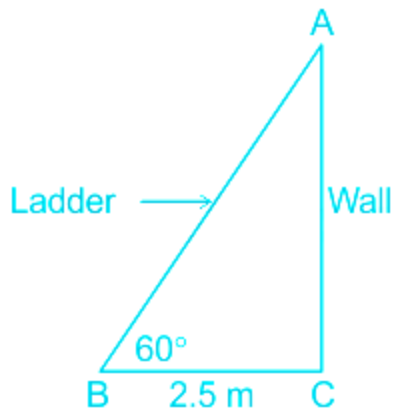
Solution:

Formula used:

$$\cos\theta = \text{Base} / \text{Hypotenuse}$$

Calculation:

Let, the length of the ladder be L



By using the above formula;

$$\cos\theta = \text{Base} / \text{Hypotenuse}$$

$$\cos 60^\circ = 2.5/L$$

$$\Rightarrow 1/2 = 2.5/L \quad (\because \cos 60^\circ = 1/2)$$

$$\Rightarrow L = 2 \times 2.5 = 5 \text{ m}$$

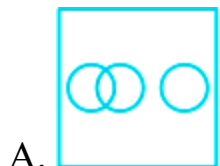
$\therefore$ The length of the ladder is 5 meters.

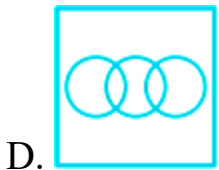
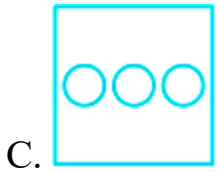
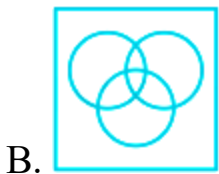
Question 36

Select the Venn diagram that best represents the relationship between the following set of classes.

Taller, Black hair people and Indians

Options:

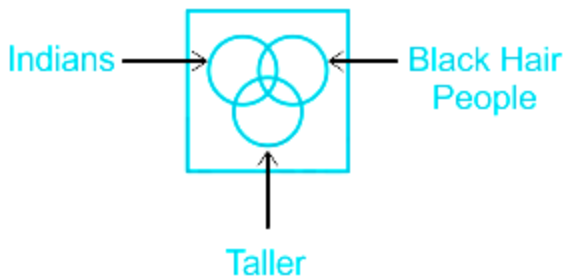




Answer: B

Solution:

The correct Venn diagram representation is



- Some Indians are Black hairpeople.
- Some Taller are Indians.
- Some Black haired people are Taller.

Hence, the correct answer is **"option (2)"**.

Question 37

The angle (in degrees) between the hands of a clock at 8:15 is

Options:

A. 125.5

B. 170.5

C. 157.5

D. 150.5

Answer: C

Solution:

Angle traced by hour hand in 8:15 $= \theta = \frac{11M}{2} - 30H$

$$\text{Angle } \theta = \frac{11 \times 15}{2} - 30 \times 8$$

$$\text{Angle } \theta = 157.5$$

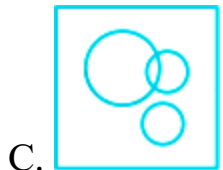
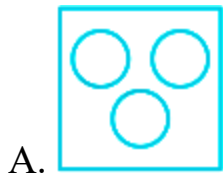
Hence, the correct answer is "option (3)".

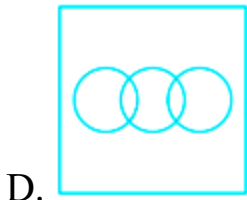
Question 38

Select the Venn diagram that best represents the relationship between the following.

Mother, Father and Engineer

Options:



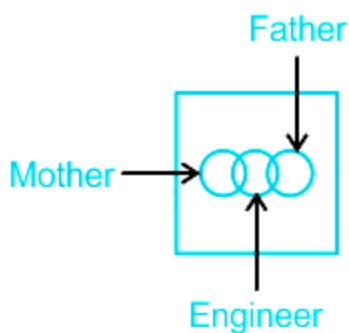


Answer: D

Solution:

The least possible Venn diagram for the given relation is a shown below :

- Some Mothers are Engineers.
- Some Fathers are Engineers.



Hence, the correct answer is "option (4)".

Question 39

Find how many meaningful word can be form, from the 1st, 4th, 9th and 10th letter of a word "CORRUPTION".

Options:

- A. One
- B. Two
- C. Three
- D. None

Answer: A

Solution:

C	O	R	R	U	P	T	I	O	N
1	2	3	4	5	6	7	8	9	10

1st letter = C

4th letter = R

9th letter = O

10th letter = N

Meaningful word = **CORN**

Therefore, only one meaningful word is formed.

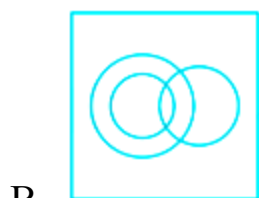
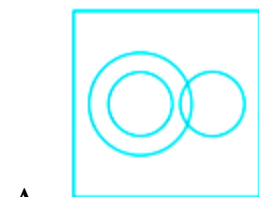
Hence, the correct answer is **"option (1)"**.

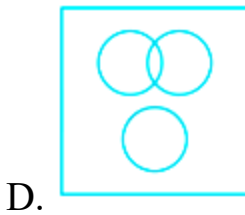
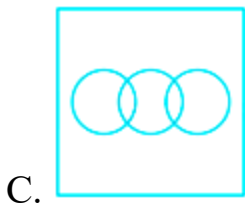
Question 40

Select the Venn diagram that best represents the relationship between the following classes.

Women, Mother, Doctor

Options:



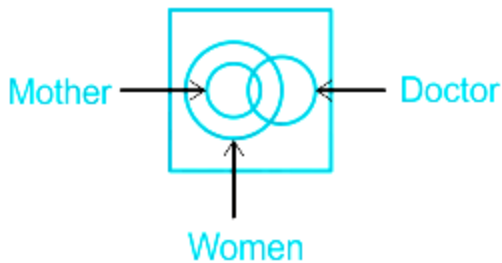


Answer: B

Solution:

The least possible Venn diagram for the given relation is a shown below :

- Some Doctors are Mothers.
- Some Women are Doctors.
- All Mothers are Women.



Hence, the correct answer is "option (2)".

Question 41

If FOSTER = 36, DANCE = 25, then ORCHARD = ?

Options:

A.

50

B.

64

C.

49

D.

67

Answer: C

Solution:

The table shows the positional value of alphabets.

Alphabets	A	B	C	D	E	F	G	H	I	J	K	L	M
Positional value	1	2	3	4	5	6	7	8	9	10	11	12	13
Positional value	26	25	24	23	22	21	20	19	18	17	16	15	14
Alphabets	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

The logic followed here is:

$$\Rightarrow \text{FOSTER} = 6 \text{ letters} = (6)^2 = 36$$

$$\Rightarrow \text{DANCE} = 5 \text{ letters} = (5)^2 = 25$$

Similarly,

$$\Rightarrow \text{ORCHARD} = 7 \text{ letters} = (7)^2 = 49$$

Hence, the correct answer is "option (3)".

Question 42

Select the number that can replace the question mark (?) in the following series.

423, 407, 439, 391, 455, ?

Options:

A. 373

B. 375

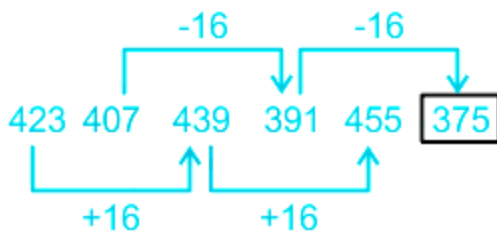
C. 374

D. 407

Answer: B

Solution:

The logic followed here is:



Hence, the correct answer is "option (2)".

Question 43

Choose the correct alternative for the following:

Light : Blind :: Noise : ?

Options:

A. Unsighted

B. Dumb

C. Deaf

D. Aphonic

Answer: C

Solution:

The logic followed here is :

A "**Blind**" cannot see the "**Lights**".

Similarly,

A "Deaf" cannot hear the "Noise".

Hence, the correct answer is "option (3)".

Additional Information

Word	Meaning
Un sighted	The one who cannot see
Dumb	The one who can not speak
Deaf	The one who can not hear the voice
Aphonic	The one who have no voice or sound

Question 44

Select the number that can replace the question mark (?) in the following series.

1, 2, 6, 15, 31, 56, ?

Options:

A. 92

B. 148

C. 90

D. 93

Answer: A

Solution:

The logic followed here is:



Hence, the correct answer is "option (1)".

Question 45

If + means $\times$, $\times$ means $-$, $-$ means $\div$, and $\div$ means $+$, then what will be the value of the given expression?

$$37 + 73 - 23 \div 32 \times 29 = ?$$

Options:

A. 89.16

B. 78.84

C. 0.09

D. 120.43

Answer: D

Solution:

B	Brackets in order {}, {}, []	ब्रैकेट {}, {}, [] क्रम में
O	of	का
D	Division ($\div$)	विभाजन ($\div$)
M	Multiplication ($\times$)	गुणा ($\times$)
A	Addition ($+$)	जोड़ ($+$)
S	Subtraction ($-$)	घटाव ($-$)

Decoding the information,

Sign	$+$	$\times$	$-$	$\div$
Means	$\times$	$-$	$\div$	$+$

Given:

$$37 + 73 - 23 \div 32 \times 29 = ?$$

After replacing the signs from left to right and using the BODMAS rule,

$$= 37 \times \underline{73 \div 23} + 32 - 29$$

$$= 37 \times 3.174 + 32 - 29$$

$$= 117.43 + 32 - 29$$

$$= 149.43 - 29$$

$$= 120.43$$

Hence, “**120.43**” is the correct answer.

Question 46

Select the figure from among the given options that can replace the question mark (?) in the following series.



Options:



Answer: D

Solution:

The pattern followed here is:

- 1st position line went to fifth position and 2nd position line is rotated 180° and moves to 1st position, 3rd position line moves to 2nd position, 4th position line moves to 3rd position and 5th position line moves to 4th position in each figure.

$\therefore$ All the line with symbols facing right side are rotated 180° and faces left side in figure 5.

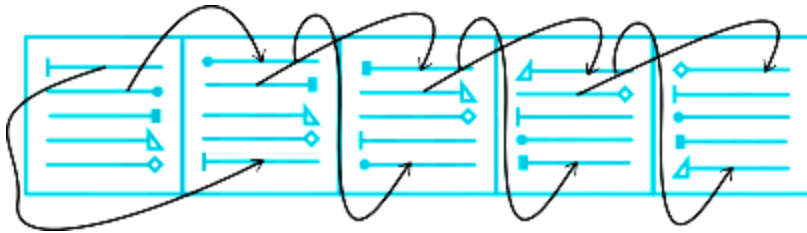


Figure 1 to figure 2 $\rightarrow$ 1st position line went to fifth position and 2nd position line is rotated 180° and moves to 1st position, 3rd position line moves to 2nd position, 4th position line moves to 3rd position and 5th position line moves to 4th position.

Figure 2 to figure 3 $\rightarrow$ 1st position line went to fifth position and 2nd position line is rotated 180° and moves to 1st position, 3rd position line moves to 2nd position, 4th position line moves to 3rd position and 5th position line moves to 4th position.

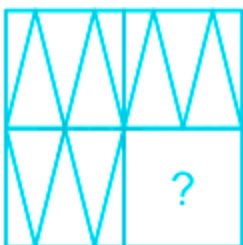
Figure 3 to figure 4 $\rightarrow$ 1st position line went to fifth position and 2nd position line is rotated 180° and moves to 1st position, 3rd position line moves to 2nd position, 4th position line moves to 3rd position and 5th position line moves to 4th position.

Figure 4 to figure 5 $\rightarrow$ 1st position line went to fifth position and 2nd position line is rotated 180° and moves to 1st position, 3rd position line moves to 2nd position, 4th position line moves to 3rd position and 5th position line moves to 4th position.

Hence, the correct answer is "**Option 4**".

Question 47

Which answer figure will complete the pattern in the following question figure?



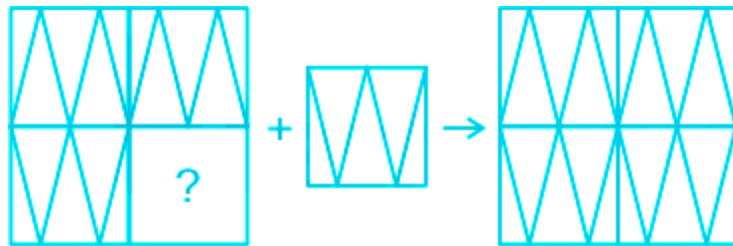
Options:



Answer: A

Solution:

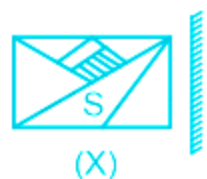
Completing the incomplete pattern:



Hence, the correct answer is "**Option 1**".

Question 48

Select the correct mirror image of figure (X) when the mirror is placed at the right side of the figure.



Options:

A.



B.



C.



D.



Answer: A

Solution:

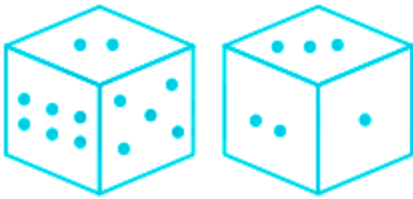
The mirror image of the question figure is as follows:



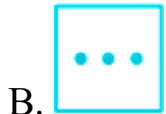
Hence, the correct answer is "**Option (1)**".

Question 49

Two different positions of the same dice are shown below, the six faces of which are numbered 1 to 6 dots. Find the number on the face opposite to the one having 2 dots.



Options:



Answer: C

Solution:

The faces with dots 5 and 6 are adjacent to the face with dots 2 in the first position

The faces with dots 1 and 3 are adjacent to the face with dots 2 in the second position.

The only number left is 4.

Hence, **4 dots (option 3)** are on the face opposite to the one having 2.

Question 50

Speaking to Imran, Salim said, "You are the Son of my only son-in-law. How is Salim wife related to Imran?"

Options:

A. Salim's wife is paternal grandmother of Imran

- B. Salim's wife is mother of Imran.
- C. Salim's wife is maternal grandmother of Imran
- D. Salim's wife is aunt of Imran.

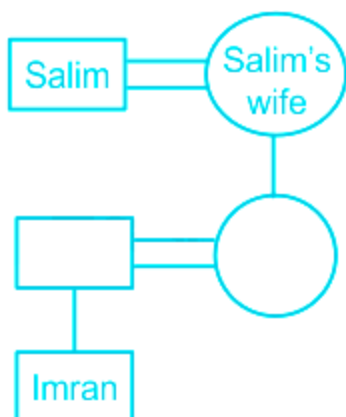
Answer: C

Solution:

Preparing the family tree using the following symbols:

Symbol in Diagram	Meaning
○	Female
□	Male
══	Married Couple
—	Siblings
	Difference of A Generation

Speaking to Imran, Salim said, "You are the Son of my only son-in-law.



As it can be seen that Imran's wife is the maternal grandmother of Salim.

Therefore, the correct answer is "**Option 3**".
